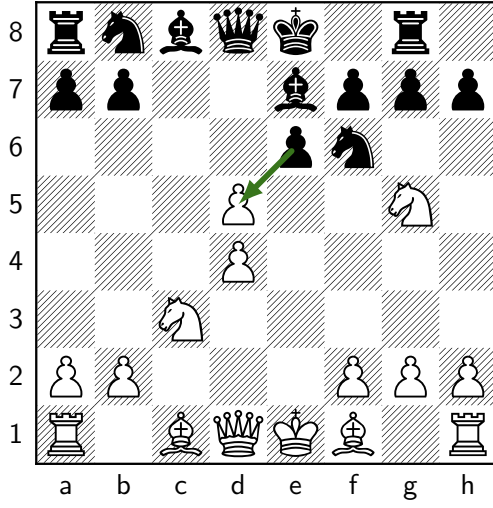


Melanie-BOTS (CCRL engines) — Caro-Kann, Panov

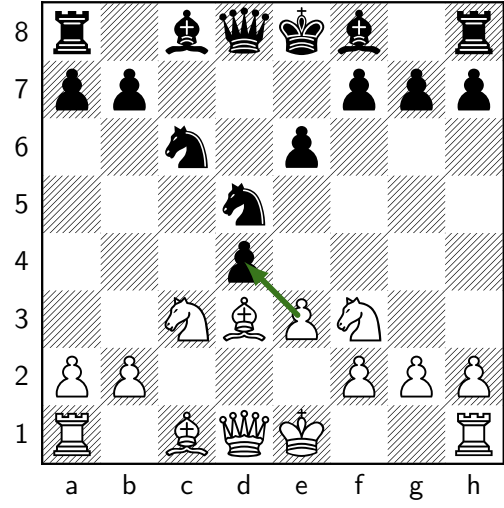
Melanie (Deep_Vision)

Abstract

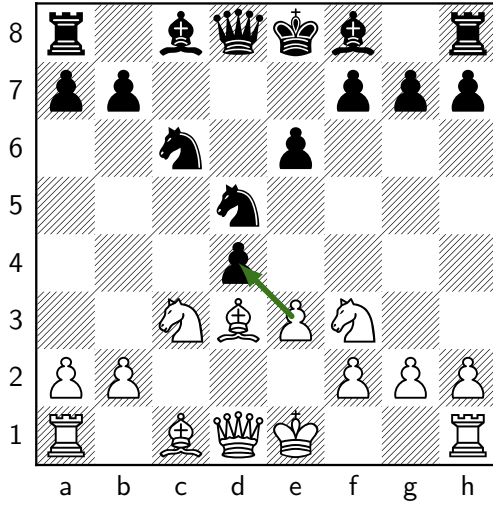
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



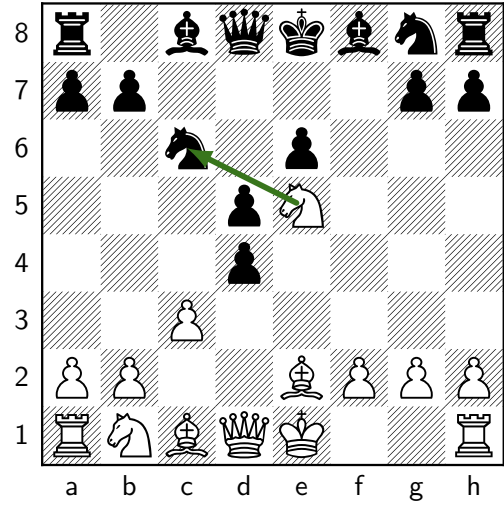
(a)



(b)

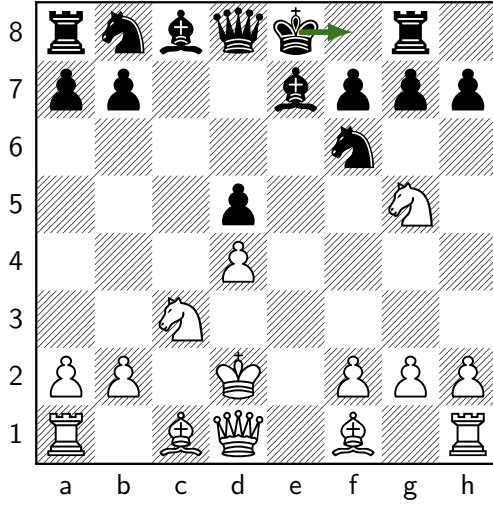


(c)

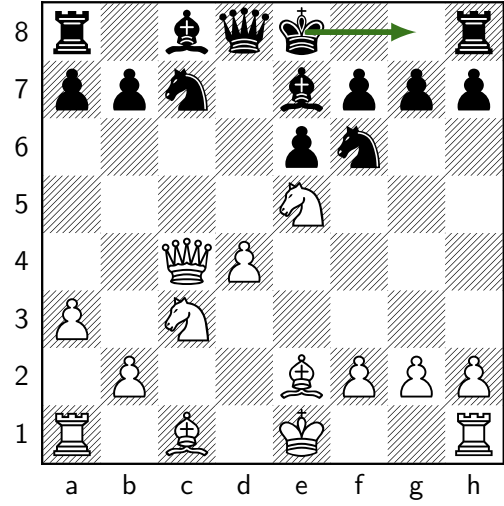


(d)

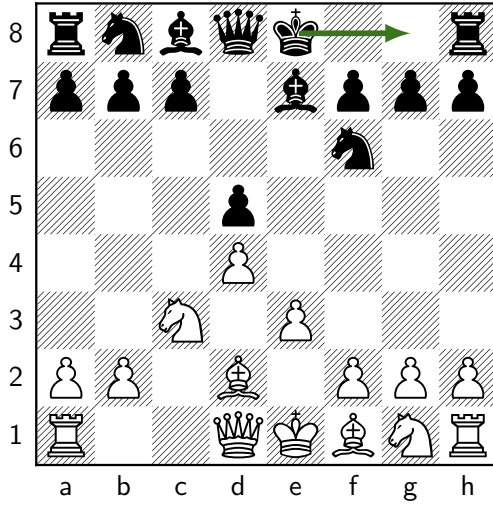
Figure 1: **Move 8, Black: exd5.** (a) Our position; Melanie plays **exd5** (**e6d5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1272` (ply 14) [structural distance $d=50$, QAP obj=58]: the master played **exd4**; under μ our *Pe3* plays the role of that game’s *Pe3*; cross-type roles: our *Pa7* as their *Rh1*, our *Nb8* as their *Pb2*; transferred move **e6d5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1273` (ply 14) [structural distance $d=50$, QAP obj=58]: the master played **exd4**; under μ our *Pe3* plays the role of that game’s *Pe3*; cross-type roles: our *Pa7* as their *Rh1*, our *Nb8* as their *Pb2*; transferred move **e6d5** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2089` (ply 14) [structural distance $d=74$, QAP obj=63]: the master played **Nxc6**; under μ our *Pe3* plays the role of that game’s *Ne5* (a **cross-type** correspondence); cross-type roles: our *Nb8* as their *Ra1*, our *Pf7* as their *Nb1*; transferred move **e6e5**. Synthesis: 2 retrieved analogs transferred to **exd5** (summed QAP objective 115, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



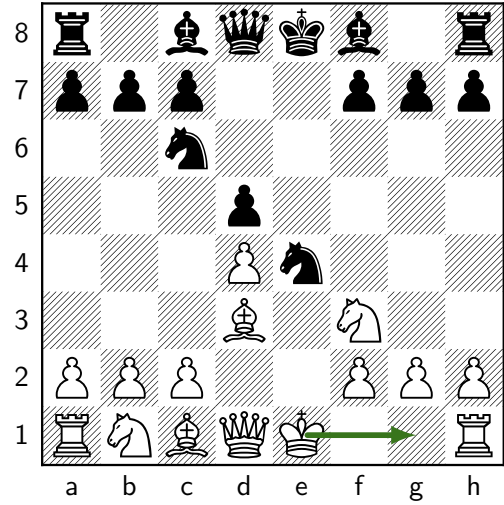
(a)



(b)

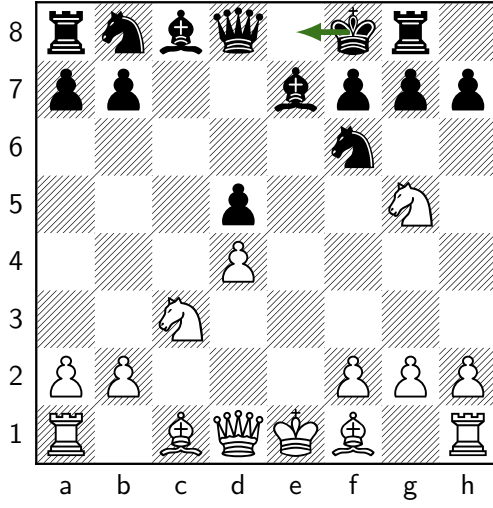


(c)

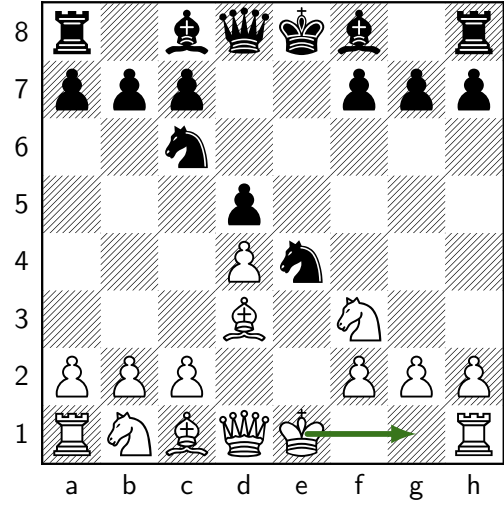


(d)

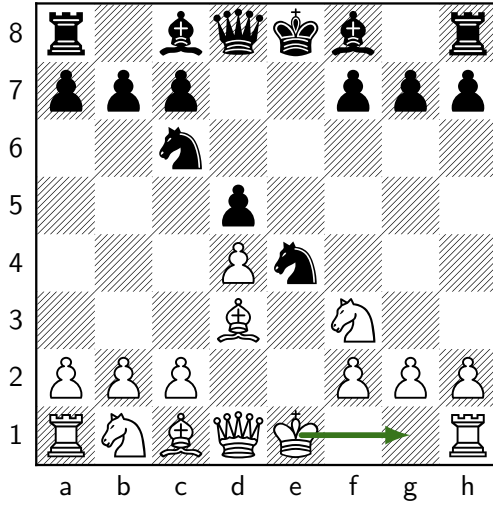
Figure 2: **Move 9, Black: Kf8**. (a) Our position; Melanie plays Kf8 (e8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3870` (ply 21) [structural distance $d=68$, QAP obj=32]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bfl* as their *Ra8*, our *pb2* as their *Bc8*; transferred move e8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2030` (ply 13) [structural distance $d=69$, QAP obj=26]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bc1* as their *Ra8*, our *pb2* as their *Bc8*; transferred move e8f8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3416` (ply 12) [structural distance $d=74$, QAP obj=6]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bfl* as their *Nb1*, our *Nf6* as their *Bc1*; transferred move e8f8 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kf8 (summed QAP objective 102, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



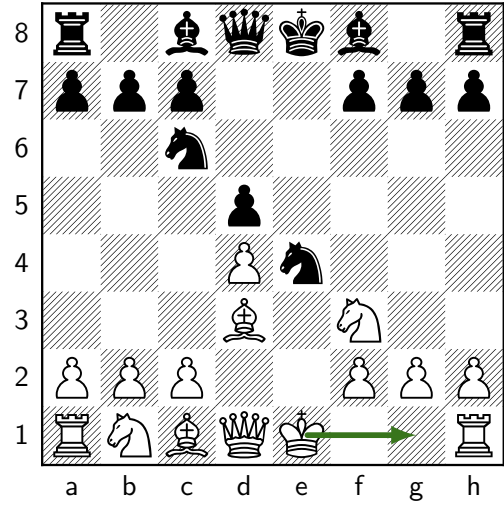
(a)



(b)

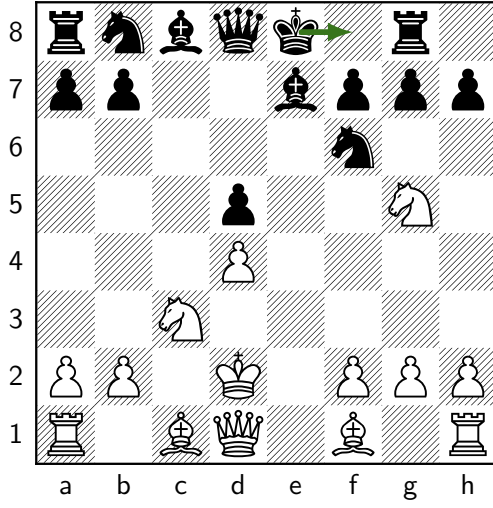


(c)

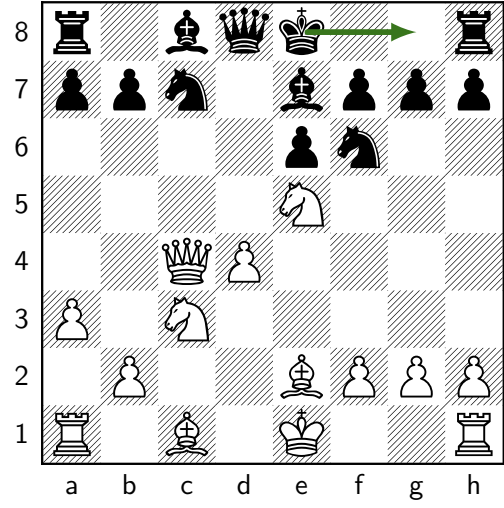


(d)

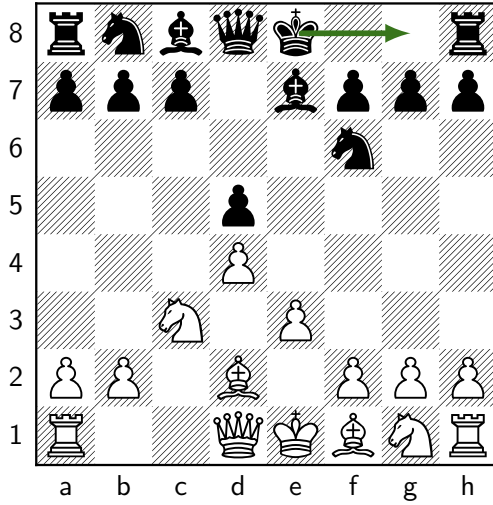
Figure 3: **Move 10, Black: Ke8.** (a) Our position; Melanie plays Ke8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3416` (ply 12) [structural distance $d=63$, QAP obj=6]: the master played 0-0; under μ our *Kf1* plays the role of that game's *Ke1*; cross-type roles: our *Pf7* as their *Nb1*, our *Nf6* as their *Bc1*; transferred move f8e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3417` (ply 12) [structural distance $d=63$, QAP obj=6]: the master played 0-0; under μ our *Kf1* plays the role of that game's *Ke1*; cross-type roles: our *Pf7* as their *Nb1*, our *Nf6* as their *Bc1*; transferred move f8e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1401` (ply 12) [structural distance $d=63$, QAP obj=6]: the master played 0-0; under μ our *Kf1* plays the role of that game's *Ke1*; cross-type roles: our *Pf7* as their *Nb1*, our *Nf6* as their *Bc1*; transferred move f8e8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke8 (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



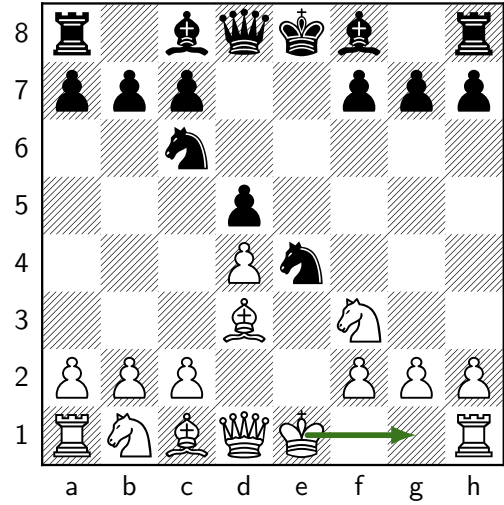
(a)



(b)

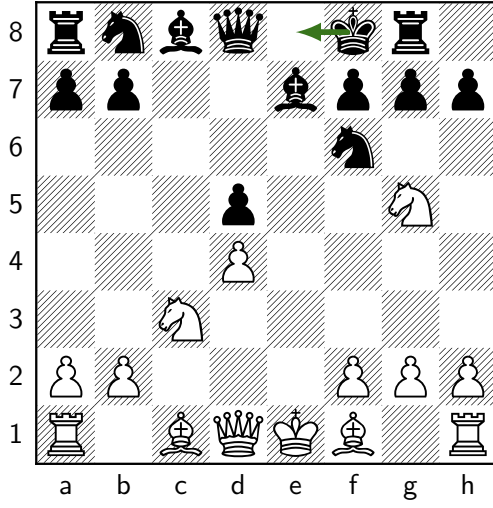


(c)

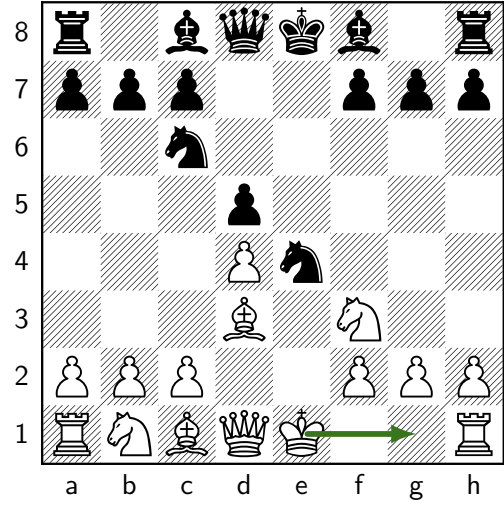


(d)

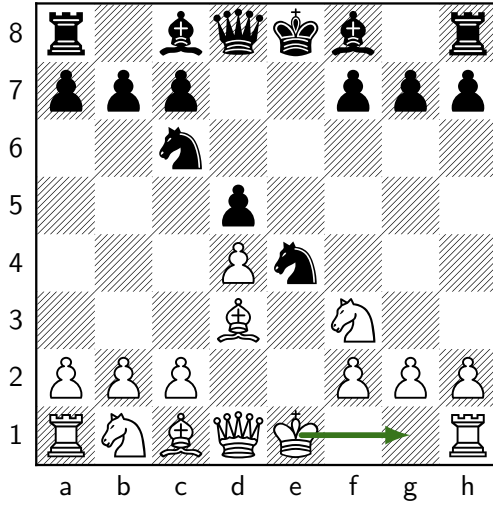
Figure 4: **Move 11, Black: Kf8**. (a) Our position; Melanie plays Kf8 (e8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3870` (ply 21) [structural distance $d=68$, QAP obj=32]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bfl* as their *Ra8*, our *pb2* as their *Bc8*; transferred move e8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2030` (ply 13) [structural distance $d=69$, QAP obj=26]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bc1* as their *Ra8*, our *pb2* as their *Bc8*; transferred move e8f8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3416` (ply 12) [structural distance $d=74$, QAP obj=6]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *bfl* as their *Nb1*, our *Nf6* as their *Bc1*; transferred move e8f8 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kf8 (summed QAP objective 102, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



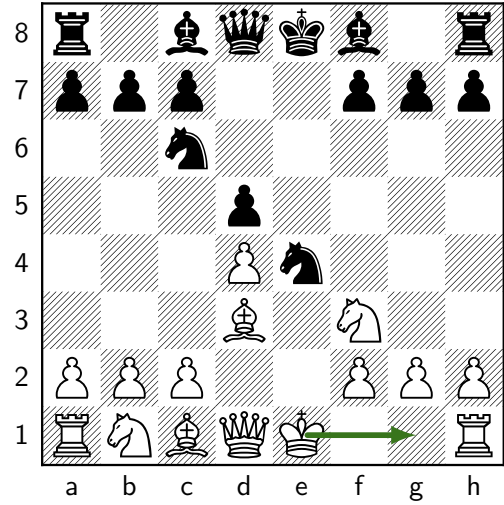
(a)



(b)



(c)



(d)

Figure 5: **Move 12, Black: Ke8.** (a) Our position; Melanie plays Ke8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3416` (ply 12) [structural distance $d=63$, QAP obj=6]: the master played 0-0; under μ our *Kf1* plays the role of that game's *Ke1*; cross-type roles: our *Pf7* as their *Nb1*, our *Nf6* as their *Bc1*; transferred move f8e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3417` (ply 12) [structural distance $d=63$, QAP obj=6]: the master played 0-0; under μ our *Kf1* plays the role of that game's *Ke1*; cross-type roles: our *Pf7* as their *Nb1*, our *Nf6* as their *Bc1*; transferred move f8e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1401` (ply 12) [structural distance $d=63$, QAP obj=6]: the master played 0-0; under μ our *Kf1* plays the role of that game's *Ke1*; cross-type roles: our *Pf7* as their *Nb1*, our *Nf6* as their *Bc1*; transferred move f8e8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke8 (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.